Morgan Stanley - Company Specific Mock Set 1 2027

Algorithm CSE | Reasoning Ability | Core Programming CSE | Verbal Ability | Databases | Quantitative Aptitude | Data Structures | Linux | Operating System

Questions

Q1 HARD | **MCQ**

Instruction:

Choose algorithms under strict constraints.

You must search frequently in a **static, sorted dataset** of 10^8 elements with **no additional memory allowed**.

A. Linear search

B. Binary search

C. HashMap

D. DFS

©Copyright MEPL 2025

Q2 HARD | **MCQ**

Instruction:

Choose the optimal approach under constraints.

You must find the longest subarray with sum ≤ K in an array of size 10^6 consisting of **non-negative integers.**

A brute-force $O(n^2)$ solution times out.

A. Sorting

B. Sliding window

C. DFS

D. Recursion

©Copyright MEPL 2025

Q3 HARD | **MCQ**

Instruction:

Evaluate constraints involving memory and repeated operations.

You must perform 10^6 repeated searches on a dataset of size 10^6. You are allowed O(n) extra memory but must minimize total runtime.

A. Linear search every time

B. Sort once, then binary search

C. Build a HashMap once and query

D. Use recursion

©Copyright MEPL 2025

Q4 HARD | **MCQ**

Instruction:

Choose algorithms under edge constraints.

You need shortest paths in a graph with negative edge weights but no negative cycles.

A. Dijkstra

B. BFS

C. Bellman-Ford

D. DFS

©Copyright MEPL 2025

Q5 HARD | **MCQ**

Instruction:

This question tests optimal algorithm selection under strict constraints.

You are given a static, sorted dataset with 100 million entries. You must perform frequent searches with minimal latency and no additional memory overhead.

Which algorithm should you use?

A. Linear Search

B. Binary Search

C. Build a HashMap

D. BFS

©Copyright MEPL 2025

Q6 HARD | **MCQ**

Instruction:

Fix exponential recursion.

A recursive Fibonacci solution fails for large inputs due to exponential time.

What is the best fix

A. Increase recursion depth

B. Use memoi

C. Use BFS

D. Sort inputs

©Copyright MEPL 2025

Q7 HARD | **MCQ**

Instruction:

Identify the logical flaw that causes incorrect window adjustment when handling repeated characters.

Which issue in the following code can cause incorrect results when a repeated character lies outside the current window?

```C++
int longestUnique(string s) {
    unordered_map<char, int> mp;
    int left = 0, maxLen = 0;

    for (int right = 0; right < s.length(); right++) {
        if (mp.find(s[right]) != mp.end()) {
            left = mp[s[right]] + 1;
        }
        mp[s[right]] = right;
        maxLen = max(maxLen, right - left + 1);
    }
    return maxLen;
}
```

A. The map stores indices instead of counts

B. The left pointer may move backward, shrinking the valid window incorrectly

C. The loop should start from index 1

D. Time complexity is O()

©Copyright MEPL 2025

Q8 HARD | **MCQ**

Instruction:

Choose the optimal approach under constraints.

You need to determine if any two numbers in an array sum to a target.

Constraints:

- Input size: 10^6
- You may modify the array
- Memory usage must remain minimal

What is the best approach?

A. Use HashMap

B. Sort array and use two pointers

C. Brute force

D. DFS

©Copyright MEPL 2025

Q9 MODERATE | **MCQ**

Instruction:

Choose the correct answer.

If in the English alphabet every fifth letter is replaced by the symbol '@', which of the following would be ninth to the left of the seventeenth element from the left?

A. H

B. @

C. M

D. G

©Copyright MEPL 2025

Q10 HARD | **MCQ**

Instruction:

Identify the precise cause of runtime failure.

Which issue causes this program to throw an exception at runtime?

```java
int[] arr = {1,2,3,4,5};
int sum = 0;

for (int i = 0; i <= arr.length; i++) {
    sum += arr[i];
}
```

A. Infinite loop due to incorrect condition

B. Accessing index equal to array length causes ArrayIndexOutOfBoundsException

C. Sum variable should be long

D. Loop should decrement instead of increment

©Copyright MEPL 2025

Q11 MODERATE | **MCQ**

Instruction:

Choose the correct answer.

In a certain code ‘METHYL’ is written as ‘UDNMXI’. In which of the following manner will the word ‘RELIEF’ be written in the same code?

A. MFSGEJ

B. MFQGDJ

C. MDSGEJ

D. MDSGDJ

©Copyright MEPL 2025

Q12 MODERATE | **MCQ**

Instruction:

Choose the correct answer.

In a certain code language, the word ‘EXTERNAL’ is written as ‘56259513’. Which of the following will be the code for the word ‘STORY’?

A. 12697

B. 102697

C. 102598

D. 12687

©Copyright MEPL 2025

Q13 MODERATE | **MCQ**

Instruction:

Direction: *A sentence is provided in the question area. One of the parts of the sentence is grammatically incorrect. Choose the incorrect part.*

Reeta, along with/her friends,/are going to Paris for a week./No Error.

A. Reeta, along with

B. her friends,

C. are going to Paris for a week

D. No Error.

©Copyright MEPL 2025

Q14 HARD | **MCQ**

Instruction:

Choose the most impactful optimization under scale.

Two large tables (10M+ rows) are joined, and only 2% of rows are relevant after filtering.

What is the best optimization?

A. Increase memory

B. Apply filtering before join

C. Add more indexes

D. Use recursion

©Copyright MEPL 2025

Q15 HARD | **MCQ**

Instruction:

Fix concurrency anomaly.

A system experiences dirty reads due to concurrent transactions.

What is the best fix?

A. Reduce indexing

B. Increase isolation level

C. Use recursion

D. Increase memory

©Copyright MEPL 2025

Q16 HARD | **MCQ**

Instruction:

Identify misuse of SQL clauses involving aggregation.

Which error prevents the following query from executing correctly?

```sql
SELECT DeptID, AVG(Salary)
FROM Employee
WHERE AVG(Salary) > 50000
GROUP BY DeptID;
```

A. GROUP BY cannot be used with AVG

B. AVG must be used only in SELECT

C. Aggregate functions cannot be used in WHERE clause

D. DeptID should not be selected

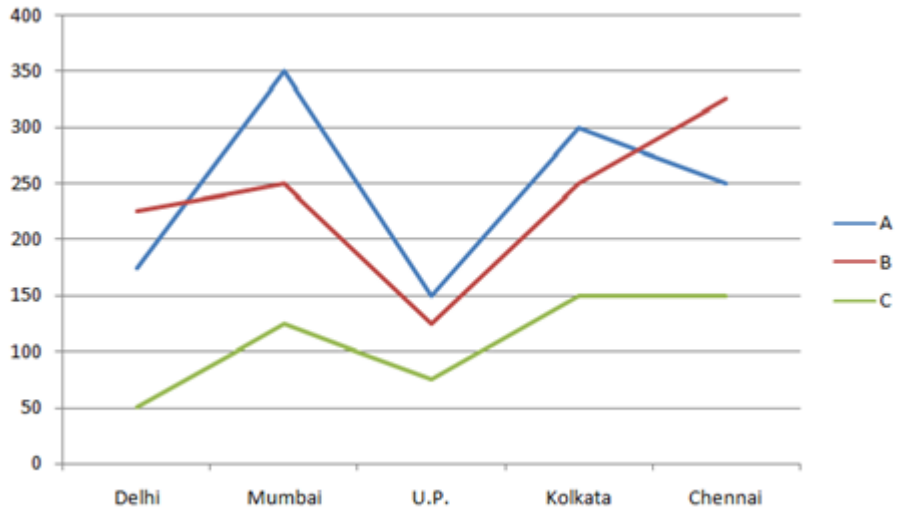
©Copyright MEPL 2025

Q17 MODERATE | **MCQ**

Instruction:

Study the following graph carefully and answer the questions that follow.

The number of passengers travelled by three different trains in five different cities:



Number of passengers travelled by train B in Chennai forms approximately what percent of the total number of passengers travelled by train B in all the cities together?

A. 23.65%

B. 25.65%

C. 29.86%

D. 27.66%

©Copyright MEPL 2025

E. 30.51%

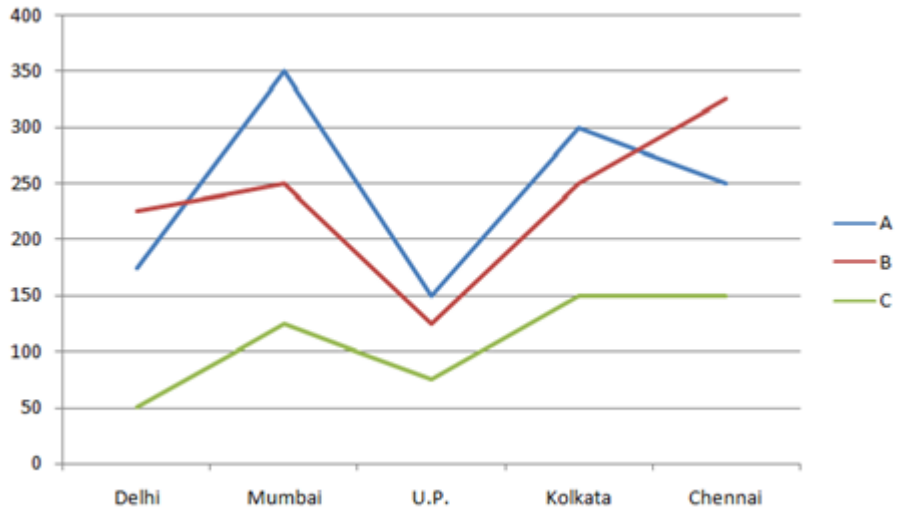
©Copyright MEPL 2025

Q18 MODERATE | **MCQ**

Instruction:

Study the following graph carefully and answer the questions that follow.

The number of passengers travelled by three different trains in five different cities:



What is the respective ratio of the number of passengers travelled in Delhi, U.P. and Chennai by all the trains together?

A. 18: 17: 29

B. 14: 18: 29

C. 15: 14: 23

D. 18: 14: 29

©Copyright MEPL 2025

E. 18: 17: 23

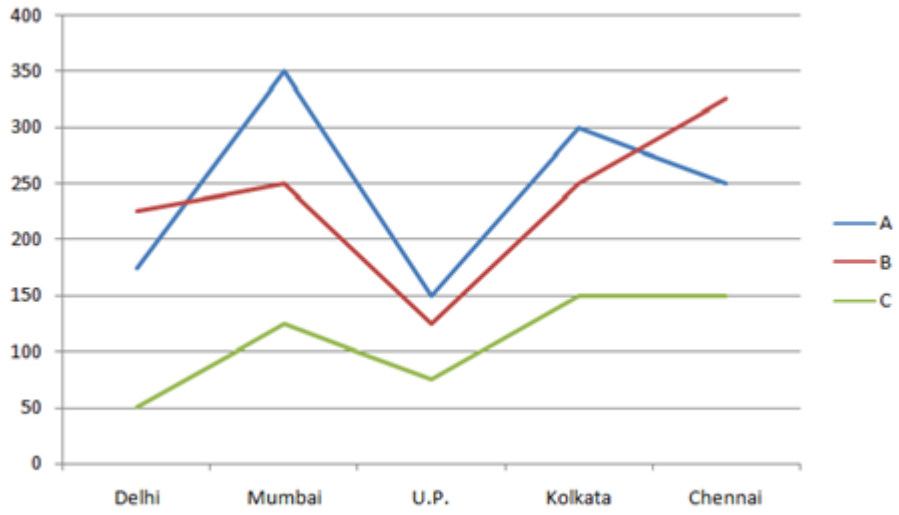
©Copyright MEPL 2025

Q19 MODERATE | **MCQ**

Instruction:

Study the following graph carefully and answer the questions that follow.

The number of passengers travelled by three different trains in five different cities:



Number of passengers travelled by train B in Kolkata is what percent of the total number of passengers travelled by all the trains in Kolkata?

A. 33.61%

B. 35.71%

C. 32.71%

D. 35.99%

©Copyright MEPL 2025

E. 34.87%

Q20 HARD | **MCQ**

Instruction:

Identify degradation under adversarial input.

A HashMap consistently shows O(n) performance under specific input patterns.

What is the most likely cause?

A. Stack overflow

B. Adversarial collisions

C. Memory leak

D. Recursion

©Copyright MEPL 2025

Q21 HARD | **MCQ**

Instruction:

Choose the optimal approach under streaming constraints.

You are receiving a continuous stream of data and must always maintain the top 100 elements seen so far.

A. Sort entire dataset repeatedly

B. Maintain a min-heap of si

C. Use a stack

D. Use DFS

Q22 HARD | **MCQ**

Instruction:

Choose structure under range query requirement.

You need sorted data with frequent updates and efficient range queries.

A. HashMap

B. Heap

C. Balanced BST

D. Stack

©Copyright MEPL 2025

Q23 HARD | **MCQ**

Instruction:

Choose the optimal data structure under strict performance guarantees.

You are designing a system where:

- Lookups must be fast
- Worst-case time must NOT degrade beyond O(log n)
- Data does not need to be sorted

Which structure should you choose?

A. HashMap

B. Balanced BST

C. Array

D. Heap

©Copyright MEPL 2025

Q24 HARD | **MCQ**

Instruction:

Choose structure under constraints.

You need:

- Frequent insertions at arbitrary positions
- Minimal memory reallocation
- No requirement for random access

Which structure is optimal?

A. Array

B. Linked List

C. Heap

D. Stack

©Copyright MEPL 2025

Q25 HARD | **MCQ**

Instruction:

Identify the pointer mismanagement that breaks traversal after reversal.

Which issue in the following linked list reversal code causes incorrect traversal?

Node* reverse(Node* head) {

```cpp
Node* reverse(Node* head) {
    Node* prev = NULL;
    Node* curr = head;

    while (curr != NULL) {
        Node* next = curr->next;
        curr->next = prev;
        prev = curr;
        curr = curr->next;
    }
    return prev;
}
```

A. prev should be initiali

B. curr should be updated using the saved next pointer

C. next pointer should be global

D. Loop should run till curr->next != NULL

©Copyright MEPL 2025

Q26 HARD | **MCQ**

Instruction:

Choose traversal under shortest-path constraint.

You must find the minimum number of steps between two nodes in an **unweighted graph**.

A. DFS

B. BFS

C. Heap

D. Recursion

©Copyright MEPL 2025

Q27 HARD | **MCQ**

Instruction:

This question tests your ability to map real-world system requirements to the most appropriate data structure.

You are building a real-time trading system where orders arrive continuously. The system must always process the highest priority order first, and priorities may change dynamically.

Which data structure should you use?

A. Queue

B. Stack

C. Max Heap (Priority Queue)

D. HashMap

©Copyright MEPL 2025

Q28 HARD | **MCQ**

Instruction:

Choose command for real-time system analysis.

A server is slow due to high CPU usage. You need a **real-time, continuously updating view** of processes.

Which command should you use?

A. ls

B. top

C. grep

D. chmod

©Copyright MEPL 2025

Q29 HARD | **MCQ**

Instruction:

Interpret permission under execution scenario.

A file has permission 750. Which users can execute it?

A. Only owner

B. Owner and group

C. Everyone

D. No one

©Copyright MEPL 2025

Q30 MODERATE | **MCQ**

Instruction:

Direction: *Identify the part of the statement that is grammatically incorrect.*

I am not sympathetic to a view that the crimes of the past can somehow be abolished by attacking and dismantling their relics.

A. I am not sympathetic to a view

B. that the crimes of the past

C. can somehow be abolished

D. by attacking and dismantling their relics.

E. No error

©Copyright MEPL 2025

Q31 HARD | **MCQ**

Instruction:

This question tests your ability to identify numeric limitations.

What is the most likely cause of incorrect results in a system using 32-bit integers when processing very large values?

A. Floating precision issue

B. Integer overflow

C. Memory leak

D. Recursion

©Copyright MEPL 2025

Q32 HARD | **MCQ**

Instruction:

This question evaluates your ability to identify root causes of failures under scale.

A backend service processes requests using in-memory data structures. Under large workloads:

Memory usage grows continuously

Garbage collection fails to reclaim memory

The system crashes with an out-of-memory error

What is the MOST likely root cause?

A. Integer overflow

B. Memory leak due to lingering object references

C. Network latency

D. Stack overflow

©Copyright MEPL 2025

Q33 HARD | **MCQ**

Instruction:

Identify the edge case that leads to incorrect or undefined behavior.

Assuming the function is expected to compute factorial for all non-negative integers, what issue exists in the following code?

```cpp
int fact(int n) {
    if (n == 1) return 1;
    return n * fact(n - 1);
}
```

A. Missing base case for n = 0

B. Function causes stack overflow for all inputs

C. Multiplication logic is incorrect

D. Recursion depth is constant

©Copyright MEPL 2025

Q34 HARD | **MCQ**

Instruction:

Identify the subtle concurrency issue that leads to inconsistent results under parallel execution.

Multiple threads call increment() concurrently, and the final value of count is inconsistent across runs. What is the root cause?

```Java
class Counter {
    int count = 0;

    public void increment() {
        count++;
    }
}
```

A. Deadlock due to missing locks

B. Race condition due to non-atomic increment operation

C. Memory leak due to shared variable

D. Stack overflow due to recursion

©Copyright MEPL 2025

Q35 EASY | **MCQ**

Instruction:

Single correct option

(N) is a whole number which when divided by 4 gives 2 as remainder. What will be the remainder when $2N^2 + 3$ is divided by 4?

A. 0

B. 1

C. 2

D. 3

©Copyright MEPL 2025

Q36 HARD | **MCQ**

Instruction:

Identify runtime behavior.

A base class reference points to a derived class object. An overridden method is invoked.

Which method executes?

A. Base class method

B. Derived class method

C. Both

D. Compile-time error

©Copyright MEPL 2025

Q37 HARD | **MCQ**

Instruction:

Identify scheduling trade-offs.

Long tasks are delayed indefinitely while short tasks complete quickly under heavy load.

Which scheduling strategy is MOST likely used?

A. FCFS

B. Round Robin

C. Shortest Job First

D. Priority with aging

©Copyright MEPL 2025

Q38 HARD | **MCQ**

Instruction:

Identify the best fix under performance constraints.

A multi-threaded application produces inconsistent results while updating shared data. Blocking must be minimized.

A. Use global lock

B. Use atomic operations or fine-grained locking

C. Increase CPU cores

D. Ignore inconsistency

©Copyright MEPL 2025

Q39 HARD | **MCQ**

Instruction:

Resolve fragmentation issue.

A system suffers from external fragmentation.

Which technique resolves this?

A. Segmentation

B. Paging

C. Stack allocation

D. Recursion

©Copyright MEPL 2025

Q40 MODERATE | **MCQ**

Instruction:

Direction: *Refer to the sentences given below. These sentences are the part of a paragraph but, have been jumbled up. Choose the correct order.*

A) uses newly designed small- molecules or enzyme complexes that can bind to the undesirable gene product or a specific RNA or RNA product and destroy it.

B) Soon you may get pills that can correct disorders in your genes.

C) One step ahead of CRISPR, the technology named RIBOTAC (ribonuclease-targeting chimeras),

D) The team says that this technology can be used to treat gene-driven diseases and even cancer.

Which of the following is the correct sequence of sentences which form a coherent paragraph?

A. CBDA

B. BCAD

C. ABCD

D. ACDB

©Copyright MEPL 2025

Q41 MODERATE | **MCQ**

Instruction:

Single Correct Option

In a class, 40% of the students like mathematics, 30% like science, and 10% like both. If a student is selected at random, find the probability that he/she likes mathematics or science?

A.

B.

C.

D.

©Copyright MEPL 2025

Q42 MODERATE | **MCQ**

Instruction:

Choose the correct answer.

A grocer mixes 26kg of wheat which cost Rs. 2 per kg with 30kg of wheat which cost Rs. 3.60 per kg and sells the mixture at Rs. 3 per kg. What is his total gain in percentage?

A. 10%

B. 8%

C. 5%

D. 7%

©Copyright MEPL 2025

Q43 MODERATE | **MCQ**

Instruction:

Direction: *Read the given passage carefully and answer the following question.*

Passage:

Automation in manufacturing, driven by robotics and artificial intelligence, has redefined industrial efficiency. Advanced robotic systems, equipped with machine learning algorithms, execute tasks with precision, reducing human error and labor costs. From automotive assembly lines to electronics production, automation enhances output while minimizing downtime. Yet, this technological shift engenders socioeconomic trade-offs. While productivity surges, automation displaces low-skill workers, exacerbating unemployment in regions dependent on traditional manufacturing. Proponents argue that automation fosters innovation, creating high-skill jobs in programming and system design. However, critics highlight the uneven distribution of these opportunities, noting that displaced workers often lack access to retraining. Furthermore, overreliance on automated systems risks vulnerability to technical failures or cyberattacks, which could disrupt global supply chains. As manufacturers navigate this landscape, balancing technological adoption with workforce resilience remains a pivotal challenge.

According to the passage, what is a primary concern associated with the widespread adoption of automation in manufacturing?

A. The potential for automation to eliminate all manufacturing jobs

B. The susceptibility of automated systems to technical disruptions

C. The inability of automation to improve production efficiency

D. The lack of precision in robotic manufacturing processes

©Copyright MEPL 2025

Q44 MODERATE | **MCQ**

Instruction:

Direction: *Read the given passage carefully and answer the following question.*

Passage:

The integration of artificial intelligence (AI) into organizational decision-making has sparked fervent debate over its ethical ramifications. Proponents argue AI enhances efficiency, leveraging vast datasets to yield data-driven insights untainted by human bias. Yet, this optimism obscures critical concerns. Algorithms, while ostensibly objective, often perpetuate biases embedded in their training data, as evidenced by recruitment tools that disproportionately favor certain demographics. Furthermore, the opacity of AI systems—often described as "black boxes"—complicates accountability when decisions adversely affect stakeholders. For instance, automated loan approval systems have been criticized for reinforcing socioeconomic disparities, raising questions about fairness. Critics also warn of over-reliance on AI, which may erode human judgment and foster a culture of deference to technology. Conversely, advocates contend that ethical frameworks, such as transparent algorithm audits and stakeholder-inclusive design, can mitigate these risks. Nonetheless, the absence of universal regulatory standards exacerbates the challenge, leaving organizations to navigate a murky ethical landscape. The tension between efficiency and equity thus remains unresolved, demanding a nuanced approach to ensure AI serves as a tool for progress rather than a perpetuator of inequity.

The primary purpose of the passage is to:

A. advocate for the widespread adoption of AI in organizational decision-making.

B. highlight the ethical challenges posed by AI in organizational contexts and suggest potential solutions.

C. argue that AI's biases are insurmountable and render it unsuitable for decision-making.

©Copyright MEPL 2025

D. compare the efficiency of AI-driven decisions with human judgment.

©Copyright MEPL 2025

Q45 MODERATE | **MCQ**

Instruction:

Direction: *Read the given passage carefully and answer the following question.*

Passage:

Cognitive biases, systematic deviations from rational judgment, profoundly influence decision-making across contexts. Confirmation bias, for instance, leads individuals to favor information aligning with preexisting beliefs, often ignoring contradictory evidence. This can distort strategic decisions in organizations, as managers may overlook market shifts that challenge their assumptions. Similarly, the anchoring effect causes over-reliance on initial information, skewing negotiations or financial forecasts. While biases are universal, their impact varies with context—high-stakes environments amplify their consequences. Behavioral economists argue that awareness and structured decision-making frameworks, such as checklists, can mitigate biases. However, critics contend that such interventions are limited, as biases are deeply ingrained and often subconscious. Moreover, cultural factors influence how biases manifest; for example, collectivist societies may exhibit stronger conformity biases. The interplay of cognition and context underscores the complexity of rational decision-making, challenging the notion that humans consistently act in their best interest. Addressing biases requires both individual vigilance and systemic safeguards.

The passage implies which of the following about cognitive biases in decision-making?

A. They can be completely eliminated through structured decision-making frameworks.

B. Their impact is uniform across all cultural and contextual settings.

C. They complicate rational decision-making and require multifaceted approaches to address.

©Copyright MEPL 2025

D. They primarily affect individual decisions rather than organizational strategies.

Q46 MODERATE | **MCQ**

Instruction:

Direction: *A, B, C, D, P, Q, R, and S are eight people seated around a square table – two on each side, facing the center.*

(i) A and B are seated at the opposite sides of the table.

(ii) R is seated immediately left to C.

(iii) Three people are seated between A and R.

(iv) D, who is sitting next to C, is facing P.

If S is seated exactly between A and D then who among the following are immediate neighbours of P.

A. Q and B

B. A and C

C. Q and A

D. Cannot be determined

©Copyright MEPL 2025

Q47 MODERATE | **MCQ**

Instruction:

Direction: *A, B, C, D, P, Q, R, and S are eight people seated around a square table – two on each side, facing the center.*

(i) A and B are seated at the opposite sides of the table.

(ii) R is seated immediately left to C.

(iii) Three people are seated between A and R.

(iv) D, who is sitting next to C, is facing P.

Who amongst the following is seated opposite to R?

A. S

B. Q

C. P

D. Cannot be determined

©Copyright MEPL 2025

Q48 MODERATE | **MCQ**

Instruction:

Direction: *In the question, given below, has three statements followed by two conclusions, numbered I and II respectively. Consider it true even if these statements seem to deviate from the commonly known facts. Pick the correct option from A, B, C, D and E.*

A) Only conclusion I follows

B) Only conclusion II follows

C) Either conclusion I or II follows

D) Neither conclusion I nor II follows

E) Both conclusion I and II follow

Which of the following conclusions follow the given statements?

Statements:

All police are racks.

All balloons are racks.

All racks are paints.

Conclusions:

I. Some paints are police.

II. Some balloons are paints.

A. A

©Copyright MEPL 2025

B. B

C. C

D. D

E. E

©Copyright MEPL 2025

Q49 MODERATE | **MCQ**

Instruction:

Choose the correct answer.

A bus left Delhi at 1 p.m. sharp. Three hours later another bus started from Delhi in the same direction. The second bus overtook the first bus at 10 p.m. Find the average speed of the two buses over this journey if the sum of their speeds is 100 km/h.

A. 50 km/h

B. 48 km/h

C. 60 km/h

D. 40 km/h

©Copyright MEPL 2025

Q50 MODERATE | **MCQ**

Instruction:

Choose the correct answer.

X and Y are two workers working in a clothing factory. X can stitch a certain number of shirts in 20 days and y in 15 days respectively. They both started the work together but after 4 days, y had to leave due to some family issues and X alone completed the remaining work.

What is the total time taken to complete the whole work?

A. 14.66 days

B. 10.66 days

C. 37.5 days

D. 25.66 days

©Copyright MEPL 2025

Answers

Q1) B	Q2) B	Q3) C	Q4) C
Q5) B	Q6) B	Q7) B	Q8) B
Q9) A	Q10) B	Q11) D	Q12) B
Q13) C	Q14) B	Q15) B	Q16) C
Q17) D	Q18) D	Q19) B	Q20) B
Q21) B	Q22) C	Q23) B	Q24) B
Q25) B	Q26) B	Q27) C	Q28) B
Q29) B	Q30) A	Q31) B	Q32) B
Q33) A	Q34) B	Q35) D	Q36) B
Q37) C	Q38) B	Q39) B	Q40) B
Q41) C	Q42) C	Q43) B	Q44) B
Q45) C	Q46) C	Q47) D	Q48) E
Q49) B	Q50) A		

Explanations

©Copyright MEPL 2025

Q1 Explanation

Binary search provides O(log n) lookup with zero extra memory.

Q2 Explanation

Sliding window works efficiently for non-negative arrays in O(n).

Q3 Explanation

HashMap enables O(1) lookup after O(n) preprocessing, optimal for repeated queries.

Q4 Explanation

Bellman-Ford handles negative weights correctly.

©Copyright MEPL 2025

Q5 Explanation

Binary search provides O(log n) lookup with zero additional memory usage.

Q6 Explanation

Memoization reduces complexity to O(n).

Q7 Explanation

If the repeated character's last occurrence is outside the current window, updating left directly can move it backward. Fix: left = max(left, mp[s[right]] + 1);

Q8 Explanation

Sorting + two pointers gives O(n log n) time and O(1) space, optimal under memory constraint.

©Copyright MEPL 2025

Q9 Explanation

According to given condition in the question, new arrangement of alphabet will be-

ABCD@FGHI@KLMN@PQRS@UVWX@Z

Now, ninth to the left of the 17th element form the left means8^{th} from the left end of English alphabet which is ‘H’.

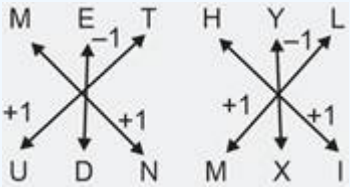
Q10 Explanation

Array indices range from 0 to length - 1. Using <= leads to invalid access.

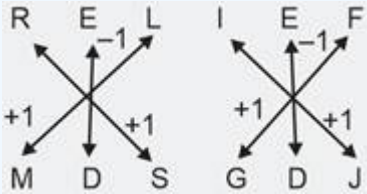
©Copyright MEPL 2025

Q11 Explanation

The given code language follows the following pattern-



Similarly,



So, 'RELIEF' is coded as 'MDSGDJ'.

Q12 Explanation

The positions of the letters of the word 'EXTERNAL', E_5 X_{24} T_{20} E_5 R_{18} N_{14} A_1 L_{12}.

Sum of the digits of Position of letters in English Alphabet is 5, (2 + 4), (2 + 0), 5, (1 + 8), (1 + 4), 1, (1 + 2) = 5,6,2,5,9,5,1,3 = 56259513

Hence, the word 'STORY' will be written as (1 + 9), (2 + 0), (1 + 5), (1 + 8), (2 + 5)

= 10,2,6,9,7 = 102697

©Copyright MEPL 2025

Q13 Explanation

Phrases like:

along with
as well as
together with
accompanied by

Do NOT make the subject plural

Correct Sentence:

Reeta, along with her friends, is going to Paris for a week.

Q14 Explanation

Reducing dataset size before join drastically improves performance.

Q15 Explanation

Higher isolation levels prevent dirty reads.

©Copyright MEPL 2025

Q16 Explanation

Aggregate conditions must be applied using HAVING, not WHERE.

Q17 Explanation

Number of passengers travelled by train B in Chennai = 325

Total number of passengers travelled by train B = 225 + 250 + 125 + 250 + 325 = 1175

Required percentage = = 27.66%

Q18 Explanation

Number of passengers travelled in Delhi by all the trains together = 175 + 225 + 50 = 450

Number of passengers travelled in U.P. by all the trains together = 150 + 125 + 75 = 350

Number of passengers travelled in Chennai by all the trains together = 250 + 325 + 150 = 725

Required ratio = 450: 350: 725 = 18: 14: 29

©Copyright MEPL 2025

Q19 Explanation

Number of passengers travelled by train B in Kolkata = 250

Number of passengers travelled by all the trains in Kolkata = 250 + 300 + 150 = 700

Required percentage = = 35.71%

Q20 Explanation

Adversarial inputs can cause many collisions, degrading performance.

Q21 Explanation

Min-heap maintains top K efficiently in O(log K) per update.

Q22 Explanation

Balanced BST supports ordered traversal and range queries.

©Copyright MEPL 2025

Q23 Explanation

Balanced BST guarantees O(log n) worst-case operations, unlike HashMap which can degrade to O(n).

Q24 Explanation

Linked list supports efficient insertions without shifting.

Q25 Explanation

After reversing the link, curr->next points backward. Traversal must continue using the stored next pointer.

©Copyright MEPL 2025

Q26 Explanation

BFS guarantees the shortest path in unweighted graphs.

Q27 Explanation

A max heap supports efficient priority updates and retrieval of the highest priority element in O(log n), making it ideal for dynamic priority systems.

Q28 Explanation

top provides real-time monitoring of processes.

Q29 Explanation

7 = rwx (owner), 5 = r-x (group), others = none.

©Copyright MEPL 2025

Q30 Explanation

the in place of a

Q31 Explanation

Values exceeding integer limits cause overflow and incorrect results.

Q32 Explanation

Unreleased references prevent garbage collection, causing memory growth and eventual failure.

Q33 Explanation

Factorial is defined for 0 as 1. Missing this base case causes incorrect behavior for n = 0.

©Copyright MEPL 2025

Q34 Explanation

count++ involves read-modify-write steps and is not atomic, causing lost updates under concurrency.

Q35 Explanation

Number N is form of 4K+2

N = 4K + 2

$2N^2 + 3 = 32 K^2 + 32 K + 11$

$\frac{32}{4}K^2 + \frac{32}{4}K + \frac{11}{4}$

Remainder = 0 + 0 + 3 = 3.

Q36 Explanation

Dynamic binding ensures derived method execution.

©Copyright MEPL 2025

Q37 Explanation

SJF favors short jobs, leading to starvation of long ones.

Q38 Explanation

Ensures correctness while minimizing contention.

Q39 Explanation

Paging eliminates external fragmentation.

Q40 Explanation

NA

©Copyright MEPL 2025

Q41 Explanation

P(Math or Science) = P(Maths) + P(Science) – P(Maths & Science)

.

Q42 Explanation

Cost of 26kg at Rs. 2 per kg = Rs. 52

Cost of 30kg at Rs. 3.60 per kg = Rs.108

∴ Cost of 56kg of the mixture = Rs.168

Selling price of 56kg = 56 × 3 = Rs.168

Total gain = 168 – 160 = 8

Gain on 160 = 8

$\frac{8\times100}{160}$ = 5%.

Q43 Explanation

The passage explicitly mentions the risk of technical failures or cyberattacks disrupting supply chains, supporting B. A is incorrect as the passage does not claim automation eliminates all jobs, only low-skill ones, while creating others. C is incorrect because the passage states automation enhances efficiency. D is incorrect as the passage notes automation's precision, not a lack thereof.

©Copyright MEPL 2025

Q44 Explanation

The passage focuses on the ethical concerns of AI in organizational decision-making (e.g., bias, opacity, accountability) while noting potential solutions like audits and inclusive design. B captures this dual focus. A is incorrect because the passage does not advocate for widespread AI adoption but critiques its challenges. C overstates the passage's stance, as it does not claim biases are insurmountable. D is incorrect, as the passage does not primarily compare efficiency but discusses ethical implications.

Q45 Explanation

The passage describes cognitive biases as complicating rational decision-making and suggests addressing them through awareness and systemic tools, implying a multifaceted approach (C). A is incorrect, as the passage notes limitations in interventions. B is incorrect, as cultural and contextual variations are mentioned. D is incorrect, as biases affect both individual and organizational decisions.

Q46 Explanation

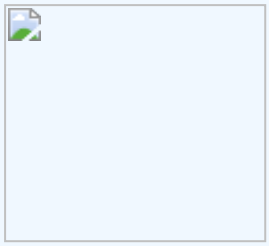
If S is seated exactly between A and D then the diagram formed will be –

From the above diagram, Q and A are immediate neighbours of P.

©Copyright MEPL 2025

Q47 Explanation

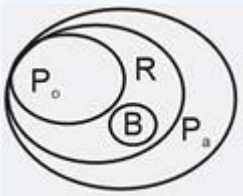
The above-given information can be represented as follows –



From the above diagram, positions of Q and S are not determined; therefore answer is cannot be determined.

Q48 Explanation

Statements given in the question can be represented as below-



It is clear from the above diagram that both conclusion follow the given statement.

©Copyright MEPL 2025

Q49 Explanation

The travel time is 9 hours & 6 hours. So the ratio of time for the travel is 9:6 i.e. 3:2 (Bus 1 to Bus 2).

Hence, the ratio of speeds would be 2 : 3. Since the sum of their average speeds is 100 km/h, therefore their respective speeds would be $\frac{2}{5}$×100 = 40 km/h and $\frac{3}{5}$×100 = 60 km/h. respectively.

Using allegation to get the answer.

Let the average speed be x.

So, $\frac{2}{3} = \frac{x-40}{60-x}$

120 – 2x = 3x – 120

5x = 240

x = 48

Hence, the average speed of two buses is 48 kmph.

Q50 Explanation

(X + Y)'s 1 day's work $= \frac{1}{20} + \frac{1}{15} = \frac{7}{60}$

work done by X and Y in 4 days $= \frac{7}{60} \times 4 = \frac{7}{15}$

Hence, remaining work $= 1 - \frac{7}{15} = \frac{8}{15}$

$\because$ work done by X in one day $= \frac{1}{20}$

$\therefore \frac{8}{15}$ work will be done by X in $20 \times \frac{8}{15} = \frac{32}{3}$ days

Hence, the total time taken $= \frac{32}{3} + 4 = 14.66$ days

©Copyright MEPL 2025